

Dear Editorial Team,

We are pleased to resubmit the revised version of our manuscript entitled "**Explainable Machine Learning Analysis of Factors Associated with Skilled Birth Attendance in Burkina Faso**" for further consideration in *Scientific Reports*.

We sincerely thank the Editor and the reviewers for their thorough evaluation and constructive comments. Their valuable suggestions have substantially improved the clarity, methodological rigor, and overall quality of our manuscript. We have carefully addressed all comments and revised the manuscript accordingly. A detailed point-by-point response letter has been provided, with all revisions referenced using the corresponding page and line numbers in the revised manuscript.

This study applies an explainable machine learning framework to identify factors associated with skilled birth attendance (SBA) using nationally representative data from 5,111 women in the 2021 Burkina Faso Demographic and Health Survey (BF-DHS). Boruta feature selection was used to identify relevant predictors, and five machine learning algorithms were developed and compared. Model interpretability was assessed using SHAP, while decision curve analysis and descriptive geographic prediction mapping were used to evaluate potential public health utility and geographic variation.

In response to the reviewers' comments, we made several important revisions, including:

- Expanded the methodological description and interpretation of model performance, adopting more appropriate language to describe the model's moderate discriminatory ability.
- Clarified that the geographic prediction maps represent **descriptive geographic variation** rather than formal spatial inference.
- Conducted additional **sensitivity analyses**, including **survey-weighted logistic regression**, **Class-Weighted Random Forest**, and **Balanced Random Forest**, to evaluate the robustness of the primary findings across different class imbalance handling strategies.
- Expanded the discussion regarding geographic variation, administrative province as a contextual predictor, health-system implications, and the policy relevance of explainable artificial intelligence (SHAP).
- Updated the manuscript, figures, tables, supplementary figures, and supplementary tables to reflect all revisions and improve transparency and reproducibility.
- Revised the Code Availability statement and updated the GitHub repository containing the analytical scripts, processed datasets, figures, and supplementary materials.

To facilitate the review process, we have included the following files:

- A clean revised manuscript (without tracked changes).
- A tracked-changes version highlighting all revisions.
- A detailed point-by-point response to reviewers.
- Updated figures, tables, supplementary figures, and supplementary tables.
- An updated GitHub repository containing all reproducible analysis materials.

We confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. The authors declare no competing interests.

We greatly appreciate the constructive feedback provided by the Editor and reviewers, which has significantly strengthened this work. We hope that the revised manuscript satisfactorily addresses all concerns, and we respectfully submit it for your further consideration.

Thank you for your time and consideration.

Sincerely,

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Response to Reviewer

Reviewer 1 – Comment 1

Comment: *The best model achieved $AUC = 0.71$, which indicates modest discrimination. The discussion occasionally implies stronger predictive performance than justified. Recommend tempering conclusions.*

Response: [page no: 2, line no: 36, page no: 12-13, line no:328-340, Pages 16, Lines 428]

We sincerely thank the reviewer for this insightful comment. We agree that an AUROC of 0.71 indicates **moderate discriminatory performance** and have revised the manuscript accordingly. Specifically, we replaced language that could overstate model performance with more appropriate terms, such as "**moderate discrimination**" and "**acceptable predictive performance**," throughout the Abstract, Results, Discussion, and Conclusion. We also clarified that, although the RF model achieved the best performance among the evaluated models, its predictive ability remains moderate and could be further improved by incorporating additional contextual and health system-related variables.

Reviewer 1 – Comment 2

Comment: *The manuscript refers to spatial inequalities, but no formal spatial model is fitted. The maps represent aggregated predictions rather than spatial inference. The wording should reflect this distinction.*

Response: [Pages 2, Lines 39-41, page no: 15, line no: 394-407, page no: 16, line no: 431-437]

We sincerely thank the reviewer for this important observation. We agree that the prediction maps represent **descriptive geographic patterns** rather than formal spatial inference. Accordingly, we revised the manuscript to consistently distinguish descriptive geographic variation from inferential spatial analysis. Specifically, we clarified in the **Methods** that the maps are descriptive visualizations based on aggregated predicted probabilities and do not involve spatial smoothing or inferential spatial modeling. We also revised the **Abstract, Results, and Discussion** to replace wording that could imply spatial inference with more appropriate terms, such as "**geographic variation**," "**geographic disparities**," and "**predicted geographic patterns**," where appropriate.

Response to Reviewer 1 – Comment 3

Comment: *The manuscript correctly evaluates on original test data. However, readers would benefit from: sensitivity analyses; class-weighted RF comparison; and Balanced RF comparison.*

Response: [Pages 10-11, Lines 273-294, page no: 16, line no: 424-427]

We sincerely thank the reviewer for this valuable suggestion. In response, we conducted additional sensitivity analyses by implementing **class-weighted Random Forest** and **Balanced Random Forest** models alongside the primary **SMOTE-based Random Forest** model. Comparative performance metrics and feature importance are presented in the **Supplementary Materials (Supplementary Table S5)**. We also expanded the **Methods** to describe these additional analyses and included a brief discussion highlighting the consistency of model performance across different class imbalance handling strategies, supporting the robustness of our findings.

Response to Reviewer

Reviewer 1 – Comment 4

Comment: *Province emerges as the strongest predictor. This raises important questions: Is province acting as a proxy for poverty? Healthcare access? Facility density? The discussion should unpack this more carefully.*

Response: [Pages 13, Lines 353-356, page no: 16, line no:431-437]

We sincerely thank the reviewer for this insightful comment. We agree that the strong contribution of the administrative province variable warrants further interpretation. Accordingly, we expanded the Discussion to clarify that province should not be interpreted as a single causal determinant; rather, it likely captures contextual differences across administrative regions, including variation in healthcare accessibility, distribution of health facilities and skilled personnel, transportation infrastructure, security conditions, and implementation of maternal health services. We also revised the SHAP interpretation to emphasize that province represents geographic contextual variation rather than an individual-level determinant.

Response to Reviewer 1 – Comment 5

Comment: *The discussion should be expanded around health-system implications, geographic inequality, and explainable AI.*

Response: [Pages 16, Lines 440-442, page no: 17, line no: 468-471, page no: 17-18, line no: 478-481]

We sincerely thank the reviewer for this valuable suggestion. In response, we expanded the Discussion to better highlight the health-system implications of our findings, emphasizing targeted interventions for low-performing provinces, including strengthened ANC services, skilled birth attendance outreach, and maternity waiting homes. We also strengthened the discussion on geographic disparities by highlighting the need for geographically targeted maternal health strategies in underserved regions. Finally, we clarified the contribution of SHAP by emphasizing its ability to improve model transparency and explain how key predictors influence SBA predictions, thereby supporting evidence-informed policy and program planning.